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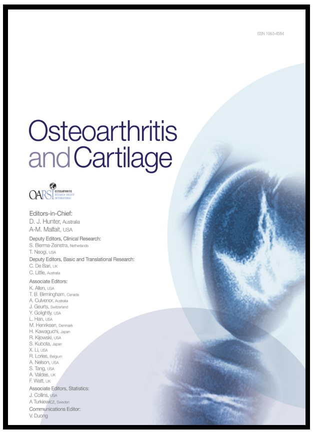

Osteocyte-Intrinsic MMP13 Exacerbates Injury-Induced Temporomandibular Joint OsteoarthritisRunning headline: Osteocytic MMP13 is required for TMJOA

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Osteocyte-Intrinsic MMP13 Exacerbates Injury-Induced Temporomandibular Joint Osteoarthritis

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ABSTRACT

Objective: Cartilage in the knee and temporomandibular joint (TMJ) rely on subchondral bone for joint homeostasis, whereby osteoarthritis (OA) causes degeneration. In the knee, suppression of osteocytic perilacunocanalicular remodeling drives cartilage degeneration in a destabilization of the medial meniscus model of OA. Given the structural and functional differences between knee and TMJ osteochondral tissues, whether osteocytes also play a causal role in TMJOA remains to be determined. We hypothesize that subchondral bone osteocytes are required for TMJ homeostasis.

Method: We evaluated the role of osteocytic MMP13 in the TMJ using 16-week-old mixed FVB and C57BL/6 male mice with osteocyte-intrinsic loss of MMP13 (DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$). DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ or Cre-negative (Control) mice received intraarticular saline or monosodium iodoacetate (MIA) to induce injury. Effects of MMP13-dependent osteocyte dysfunction on TMJ or knee OA were determined by histology and microCT.

Results: Unlike baseline differences in knees of DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ and Controls, osteocytic MMP13-deficiency had no discernible effects on TMJ cartilage and subchondral bone in the absence of injury. Relative to saline Control, MIA Control mice had cartilage degeneration, with higher modified TMJ Mankin scores (p<0.001; 95%CI [-7.7 to -1.9]). MIA-injured Controls also had increased osteocyte lacunar number (p=0.03; 95% CI [-394.2 to -14.8]) and a genotype and treatment interaction effect on TRAP stain (p=0.0213;36.04% of total variation). Contrary to our hypothesis, TMJs of MIA-injured DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice were protected from cartilage degeneration or subchondral bone changes. Relative to Cre-

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negative Controls, DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ knees showed resistance to injury-induced increase in subchondral bone plate thickness.

Conclusion: Despite different roles for osteocytic MMP13 in TMJ and knee, osteocytic MMP13 deficiency protects both joints from injury-dependent effects on subchondral bone.

Keywords: Osteocyte, osteoarthritis, temporomandibular joint, perilacuno canalicular remodeling, monosodium iodoacetate, pathogenesis

Running headline: Osteocytic MMP13 is required for TMJOA

INTRODUCTION

Temporomandibular joint osteoarthritis (TMJOA) is a painful condition associated with degeneration of cartilage and other TMJ tissues and often are part of a spectrum of temporomandibular disorders (TMDs) that are strongly linked to comorbidities such as chronic pain, fibromyalgia, depression, anxiety, and suicidal tendencies[1, 2]. The lack of effective treatments arises from a limited understanding of TMJOA mechanisms and etiopathogenesis. Given our work and that of others[3-7] demonstrating the key role of subchondral bone and its component cells, particularly osteocytes in the contributing to OA in appendicular joints via osteochondral interactions, here we investigate the role of subchondral bone in TMJ degeneration, with a focus on osteocytes.

Osteocytes are terminally differentiated cells derived from osteoblasts. Embedded within the mineralized bone extracellular matrix (ECM), osteocytes extend dendrites through a prolific lacunocanalicular network (LCN) to retain connectivity with one another, other cell types, and the vasculature[8]. Since osteocytes comprise >90% of bone cells[9], the surrounding LCN represents an enormous surface area that osteocytes actively maintain through the process of perilacunocanalicular remodeling (PLR)[10]. During PLR, osteocytes secrete proteases, such as MMP2, 13, and 14 and cathepsin K, and acidify their microenvironment to drive resorption of

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the local bone ECM, which can later be replaced[11-15]. Through mechanisms that remain incompletely understood, suppression of PLR by aging[16], glucocorticoid treatment[1, 17], or other pharmacologic or genetic interventions[18-20], can compromise osteocyte dendricity, LCN and collagen organization, mineralization and material quality of bone ECM. PLR suppression is sufficient to cause appendicular and craniofacial bone fragility, independently of bone mass. Therefore, control of bone quality through osteocytic PLR complements the control of bone quantity by osteoblasts and osteoclasts to support the mechanical integrity of the skeleton.

In addition to a crucial role in bone, osteocytic PLR also supports joint homeostasis. Subchondral bone of osteoarthritic human tibial plateaus possesses several hallmarks of PLR suppression that are not observed in non-arthritic knees[3]. Furthermore, osteocyte-intrinsic deletion of MMP13, a critical enzyme required for PLR[12], suppresses PLR in subchondral bone and drives degeneration of articular cartilage of the knee[3]. This study implicates osteocytes in bone-cartilage crosstalk and suggests that changes in subchondral bone may precede articular cartilage degradation[3, 12]. However, whether subchondral bone osteocytes also support homeostasis and contribute to progression of disease in the TMJ that has distinct differences in tissue composition, organization and function than the knee joint remains to be determined.

Here we hypothesis that osteocytic PLR in subchondral bone is required for TMJ homeostasis as it is in the knee joint. To test this hypothesis, we evaluated the TMJ radiographically and histologically in mice with an osteocyte-intrinsic deficit in MMP13, under the DMP1-cre promoter, DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ or in their Cre-negative littermate controls. The role of subchondral osteocyte function in response to sham or OA-inducing conditions was compared between the TMJ and the knee. This study expands upon the understanding of osteocytic mechanisms that are shared and those that distinguish OA progression in the TMJ and the knee, highlighting the need to consider joint-specific factors in the development of effective OA therapies.

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METHODS

Mouse Studies

We generated and characterized mice with MMP13-deficient osteocytes as previously described[3]. Briefly, homozygous MMP13$^{fl/fl}$ mice (JAX stock #005710), on an FVB background, containing loxP sites flanking MMP13 exons 3-5, were mated with hemizygous 9.6kb DMP1-cre$^{+/-}$ mice (JAX stock#023047), on a C57BL/6 background, which express Cre recombinase primarily in osteocytes and odontoblasts[21, 22]. Mice backcrossed at least 7 generations were used to generate DMP-Cre$^{-/-}$;MMP13$^{fl/fl}$ (Control) and DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice, genotypes that were confirmed with PCR of tail biopsy DNA. Mice were housed in groups in a pathogen-free environment with temperature range between 67°F-74°F, humidity between 30-70%, 12-hr light/dark cycle, free access to water and irradiated standard chow (LabDiet 5058-PicoLab Rodent Diet 20) ad libitum. Animal experiments were approved by the Institutional Animal Care and Use Committee (IACUC) at the University of California, San Francisco.

To chemically induce TMJOA, we used an established model of administering monosodium iodoacetate (MIA) known to induce TMJOA within 4-weeks post-MIA treatment on rats[23]. Briefly, 12-week-old DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ and littermate controls (n=6-9 mice/group) male mice were stratified by genotype and blindly assigned randomly for administration of a 10μL intra-articular TMJ injection of either saline or 0.10mg (4mg/kg) of MIA (Sigma I2512) dissolved in saline, as described[23]. Briefly, a 25-guage 0.625-inch needle was placed parallel to the mouse head in a coronal plane and inserted parallel to and directly inferior to the zygomatic arch and lateral to the mouse molars. At 4 weeks post-injection, mice were euthanized at skeletally mature 16-weeks of age for collection and analysis of cranial tissues

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including the mandible. Post-surgical care, including soft diet, approved by IACUC was followed.

OA of the knee was induced using the established meniscal-ligamentous injury (MLI) model[24], which results in OA by 8-weeks post-injury. Briefly, 8-week-old male DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice and littermate controls were randomly assigned into either sham or MLI group (n=6-11 mice/group). Bilateral MLI surgery was performed on anesthetized mice as described[3, 4]. Briefly, under sterile conditions, once an incision of the skin and joint capsule medial to the patella was made, the medial collateral ligament was transected, and the medial meniscus was removed. Sham mice received bilateral incision of the skin without joint capsule disruption, medial collateral ligament transection, or meniscus removal. All incisions were closed with suture and post-surgical care approved by IACUC was followed. Mice were euthanized 8-weeks post-injury at skeletally mature 16-weeks of age for collection and analysis of hindlimbs.

The sample size of 7 specimens per group was determined by power calculation with alpha=0.05 (2-sided), beta=0.20 an 80% chance of detecting a significant difference ($p<0.05$); E=0.22-.10=0.12; S=0.08, q1=0.5, q0=0.5, and previous studies[3, 23]. Sample size for each assay described below was determined based on previous studies[3, 4]. The subset of samples allocated to each assay was chosen randomly.

Histology

The right half of the cranium with right hemi-mandible and TMJ, and the right hindlegs with intact knee joints, were dissected and fixed in 10% neutral buffered formalin (NBF), decalcified in 10% EDTA, and subjected to serial ethanol dehydration and paraffin embedding. The right knee joints were positioned at a 90-degree of flexion in paraffin as previously described[3, 4]. For the TMJ, the right half of the skull was trimmed parallel to the nasal frontal bone axis along the orbital bone embedded with the joint flush with the cutting surface. Coronal sections of the

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trimmed right half of the skull and knee joints were generated using a microtome (Leica Microsystems, Buffalo Grove, IL) at 6-7μm thick, followed by standard dewaxing and hydration protocols prior to histological analyses described below. All brightfield images were obtained on a Nikon Eclipse E800 microscope.

Safranin-O/Fast Green Stain and OA Scoring

For histological assessment of OA, paraffin sections were stained with Safranin-O/Fast Green as previously described[3, 4] using a protocol adapted from the University of Rochester[25] that incorporates the following modifications: Weigert's Iron Hematoxylin for 5 minutes, brief rinse in water, differentiation in 1% acid-alcohol solution (1% hydrochloric acid in 70% ethanol), incubation in 0.02% Fast Green for 15 minutes, and staining with 1.0% Safranin-O for 20 minutes. Stained slides were subsequently dehydrated, cleared, and mounted.

For TMJOA scoring, Safranin O/Fast Green-stained sections (n=5-6/group, based on previous studies[12, 26]) were evaluated using an adaptation to the Modified Mankin Score by three blinded graders. The Modified TMJ Mankin Score was adapted to include additional metrics of subchondral bone, as described in Table 1 and Supplementary Figure 1[27]. The mean score across all graders was calculated for each sample and then averaged within each experimental group.

Ploton Silver Nitrate Stain

The LCN of the TMJ condyle subchondral bone and the tibial plateau subchondral bone was visualized in paraffin sections (n=5-6/group) stained with Ploton Silver Nitrate stain as previously described[3, 28, 29]. Briefly, sections were stained in two parts 50% silver nitrate and one part 1% formic acid in 2% gelatin solution for 55 minutes in the dark. Quantitative analysis of the subchondral bone was performed using three high-resolution images (100X) of each TMJ (n=5-6/group, based on previous studies[4, 12, 26]), and three high-resolution images (100X) on each the medial and the lateral side of the tibial plateau (n=5/group). A blinded grader manually counted lacunae using ImageJ cell counter within a manually contoured bone area to calculate the number of lacunae per total bone area. The mean number of lacunae over total bone area was averaged for each animal and then averaged within each experimental group.

Analysis of Collagen Fiber Orientation by Picrosirius Red Stain

Polarized light microscopy was conducted on paraffin-embedded sections of the TMJ, stained with a saturated aqueous solution containing picric acid (Cat# P6744, Sigma-Aldrich, St. Louis, MO) and 0.1% Sirius red (Cat# 365548, Sigma-Aldrich, St. Louis, MO) as previously described[30]. During microscopy, polarized filters were adjusted to maximize birefringence before capturing images. In a uniform region of interest in the middle of the subchondral TMJ (n=5/group, based on previous studies[12]), birefringence was quantified using OrientationJ, an ImageJ plug-in, by blinded graders, as described[31]. For each sample, collagen orientation angles were graphed with the mean maximum peak value set to 0 degrees, and the full width at half maximum (FWHM) was calculated.

Tartrate-resistant acid phosphatase stain

The tartrate-resistant acid phosphatase (TRAP) Leukocyte Acid Phosphatase staining kit (Sigma cat#387) was used to evaluate bone resorption activity in the TMJ with the following slight modifications to the manufacturer's instructions as previously described[26]. Briefly, deparaffinized and rehydrated sections were post-fixed for 30 seconds in Fixative Solution, rinsed in water, and incubated with a mixture of Fast Red Violet (Sigma cat#F3381) and Fast Garnet GBC Base Solution for 1 hour and 15 minutes at 37ºC in the dark. Stained slides were rinsed in water and counterstained with 0.02% Fast Green (Sigma cat# F3381) and mounted. Bright field 20X images (n=1-3 images/TMJ; n=4 TMJs/group, based on previous studies[12,

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[26]), were analyzed by a blinded grader using TrapHisto Software analysis to assess percent Osteoclast Surface over Bone Surface (Oc.S/BS)[32].

Immunohistochemistry (IHC)/Immunofluorescence (IF) analysis of PLR enzyme expression

Immunohistochemistry (IHC) was performed on paraffin sections using Innovex Universal Animal Immunohistochemistry Kit (329ANK) to spatially examine Cathepsin K (CTSK) expression and MMP13 expression was evaluated using immunofluorescence (IF). For IHC, after slides were deparaffinized and rehydrated, blocking with Fc-Block and Background Buster for 45 minutes each at room temperature was completed prior to incubation with primary antibody in PBS (anti-CTSK, 1:75, Abcam ab19027) at 4°C overnight. Secondary linking antibody and HRP-enzyme incubations were performed for 10 minutes each at room temperature, followed by incubation with fresh DAB solution for 5 minutes at room temperature. Slides were mounted with Innovex Advantage aqueous mounting medium.

Using 40X images of the subchondral bone (n=2-3 images/sample) and cartilage (n=4-6 images/sample) per sample (n=5/group, based on previous studies[3, 4, 12, 18, 33]), a blinded grader thresholded images in ImageJ to determine the % positive stain relative to tissue area. The average % positive stain relative to tissue area was averaged for each animal and then averaged within each experimental group.

For IF, deparaffinized and rehydrated sections were incubated with Unitrieve at 60°C for 30 minutes for antigen retrieval, rinsed in water, blocked with Background Buster at room temperature for 1 hour, incubated with PBS/0.1% Tween for 5 minutes, and incubated with primary antibody in PBS/0.1%Tween/5%BSA (anti-MMP13 1.67:100, Proteintech 18165-1-AP or Rabbit IgG 0.6:100, Abcam ab172730) overnight. The next day, after washes in PBS, secondary goat anti-rabbit antibody conjugated to Alexa Fluor Plus 594 (1:1000, Invitrogen

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A32740) was incubated for 1 hour, rinsed in PBS washes, and mounted with Prolong Gold antifade with DAPI.

Immunofluorescence images were obtained on a Leica DM2500B with X-cite 120 LED microscope using standardized exposure and brightness for all 40X images (n=4-6 images/TMJ, with n=4-6 TMJs/treatment group, based on previous studies[3, 4, 12, 18, 33]). Percent MMP13-positive stain was quantified by a blinded grader using ImageJ by thresholding gray-scale images derived from red-channel images, after normalizing to average percent MMP13-positive stain in Rabbit IgG-stained samples.

Micro-computed tomography (μCT) analysis of subchondral bone

The TMJ skeletal phenotype was analyzed by micro-computed tomography. The left TMJ was fixed in 10% NBF and stored in 70% ethanol for scanning at the University of Michigan School of Dentistry's μCT Core (funded in part by NIH/NCRR S10RR026475-01). Specimens were placed in a 19mm diameter holder and scanned over the entire width of the mandible using a μCT system (μCT100 Scanco Medical, Bassersdorf, Switzerland). Scan settings were as follows: voxel size 12μm, 70kVp, 114μA, 0.5mm AL filter, and an integration time of 500ms. Analysis was performed by a blinded grader using the manufacturer's software to isolate the articular surface of the condyle, and a fixed global threshold of 20% (200 on a grayscale of 0–1000) was used to segment bone from non-bone (n=3-4/group, based on a previous study[23]).

The knee skeletal phenotype was analyzed by scanning the left femur at the University of California, San Francisco Core Center for Musculoskeletal Biology and Medicine Skeletal Biology and Biomechanics Core using a Scanco μCT50 specimen scanner. A 4-mm region was scanned with the following settings: voxel size 10um, 55kVp, 109μA, and integration time of 500ms. Femoral subchondral bone plate (SBP) thickness was contoured by a blinded reviewer

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in sagittal μCT images as described[4]. Briefly, circles of diameter 750μm were placed on a grayscale 3D thickness map on the center of each knee quadrant (lateral anterior, medial anterior, lateral posterior, medial posterior), and mean pixel intensity was measured and converted to SBP thickness in μm (n=10 mice/group, based on previous studies[4, 12, 26]). Representative grayscale images were converted to pseudocolor in the figures.

Statistical Analysis and Data Management

Sample size was determined by power calculation with alpha=0.05 (2-sided), beta=0.20 an 80% chance of detecting a significant difference ($p<0.05$); E=0.22-.10=0.12; S=0.08, q1=0.5, q0=0.5, and previous studies[3, 23]. Comparisons between two groups were tested with unpaired two-tailed Student's *t*-test. Comparisons between MIA-treated/MLI-injury and genotype in mice were tested with two-way ANOVA with post-hoc Tukey using GraphPad Prism (GraphPad Software version 10) and statistical significance required a $p<0.05$ compared with respective controls. All data are represented as mean + standard deviation (SD). The number of samples per group is denoted as "n". The full report of the statistical analyses for each assay is in the Supplementary Excel file.

RESULTS

Establishment of a Monosodium Iodoacetate (MIA) Injury Model to Induce TMJOA in Mice

MIA as a model of inducing OA has been previously established in the knee and TMJ of rats and mice[34, 35 23, 36] and has been used to study pain and behavior outcomes associated with OA as this model mimics OA pain in humans[37]. MIA blocks glycolysis by inhibiting glyceraldehyde-3-phosphate dehydrogenase[38] and can induce several hallmarks of OA, including chondrocyte death, cartilage degeneration, osteophyte formation, and synovial hyperplasia[39]. Therefore, we used the MIA model to study the role of osteocytes in mouse

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TMJOA. After pilot studies testing MIA dose and duration in mice (data not shown), we found that 4-weeks post-injection of 0.10mg of MIA provides reproducible results and is sufficient to induce cartilage loss that mimics TMJOA as assessed in Safranin-O/Fast Green staining (Fig.1). TMJOA was graded histologically using a Modified TMJ Mankin score[27] that incorporates 2 additional metrics, subchondral bone deterioration and demarcation between cartilage and subchondral bone that are specific to the TMJ (Table 1, Supplemental Figure 1).

Protective Role of Osteocytic MMP13-Deficiency against MIA-Induced TMJOA

Intra-articular administration of MIA induced TMJ degeneration in Cre-negative control mice (Fig.1), which showed significantly higher scores for loss of background staining ($p=0.02$; 95%CI [-2.8 to -0.2]) (Fig.1C), poorly demarcated cartilage ($p=0.001$; 95%CI [-1.3 to -0.3]) (Fig.1D), enhanced chondrocyte periphery staining ($p=0.02$; 95%CI [-1.4 to -0.1]) (Fig.1E), and altered chondrocyte spatial arrangement with diffuse hypocellularity ($p<0.001$; 95%CI [-2.9 to -0.8]) (Fig.1F) relative to saline controls. Collectively, these changes contribute to an overall increase in the Modified TMJ Mankin score in the MIA-treated control group compared to saline-treated controls ($p<0.001$; 95%CI [-7.7 to -1.9]) (Fig.1B), confirming that MIA injection reproducibly and robustly induces TMJOA.

The TMJs of saline-injected DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice did not show any differences in histologic features or any Modified TMJ Mankin score parameters from saline-injected Cre-negative controls. The lack of a baseline TMJOA phenotype in DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice was unexpected, since DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice show baseline OA of the knee relative to their Cre-negative controls[3]. Furthermore, unexpectedly, MIA treatment did not induce OA in the TMJ of DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice, as it does in controls. Neither the overall Modified TMJ Mankin scores of any of its component subcategories showed significant effects of MIA, relative to saline-injection, on the TMJ of DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice (Fig.1B-F). This,

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together with the OA phenotypic changes modulated by MIA in Cre-negative mice, suggests that reduction of osteocytic MMP13 protects the TMJ from MIA-induced OA.

Osteocytic MMP13-Deficiency Prevents Injury-induced Changes in Osteocyte Density but Collagen Organization Remains Unchanged

Since changes in subchondral bone can precede changes in the cartilage[3, 4], we examined the LCN in the subchondral region of the TMJ histologically using Ploton silver stain (Fig.2A). Relative to saline-injected TMJs, MIA administrations significantly increased the number of lacunae per bone area (p=0.03; 95%CI [-394.2 to -14.8]) (Fig.2B) in Cre-negative controls, suggesting that MIA treatment alters LCN homeostasis. In contrast, MIA-treated DMP1-cre$^{+/-}$;MMP13fl/fl mice did not change osteocyte density (Fig.2B), suggesting that the reduction in osteocyte-intrinsic MMP13 protects LCN homeostasis from the adverse effects of MIA-treatment.

Since osteocyte dysfunction and PLR suppression can be associated with collagen disorganization, including in DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ long bone[3], we evaluated the structural orientation of the collagen fibers within the TMJ condyle in picrosirius red stained sections (Fig.2C). Relative to the saline-treated control TMJ, we identified a nonsignificant trend of decreased collagen organization in MIA-treated control groups (Fig.2E). Collagen organization (Fig.2D), maximum distribution of orientation (Fig.2E), and full width half maximum (FWHW) (Fig.2F) were largely unchanged across all groups. These results were surprising since disruptive collagen organization was observed in knees from DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice[3], indicating that while changes in collagen orientation is highly dynamic, osteocytic MMP13's role in regulating collagen organization is also region dependent. Cumulatively, these findings show that osteocytic MMP13-deficiency protects the TMJ from adverse effects of MIA on cartilage and subchondral bone LCN, but collagen organization remains unaffected.

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Maintenance of Subchondral Bone Remodeling Following Injury in the DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ TMJ is Independent of Cathepsin K.

During PLR, osteocytes secrete enzymes such as MMPs and cathepsin K (CTSK) to execute local resorption of the perilacunar and pericanalicular bone matrix. Given that osteocyte density in the DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ TMJ subchondral bone was unaltered by MIA, we examined bone resorption activity via TRAP stain (Fig.3A) in conjunction with the expression of PLR enzyme CTSK using IHC (Fig.3D). A modestly elevated TRAP staining and percent osteoclast surface over bone surface (Oc.S/BS) appeared in the TMJ subchondral regions of MIA-injected controls compared to saline-treated Cre-negative controls (Fig.3A-B). Similarly, relative to saline-treated Cre-negative controls, MIA-treated controls showed a trending increase in percent Oc.S/BS throughout the TMJ, and a nonsignificant increase in CTSK-positive cells in both the TMJ articular cartilage and subchondral bone (Fig. 3D-G). This mildly elevated CTSK and elevated bone resorption upon MIA-treatment may contribute to the TMJOA progression as observed with elevated Modified Mankin Score.

Basal DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice also had a nonsignificant increase in % Oc.S/BS (Fig.3A-B) and unaltered CTSK-positive stain (Fig.3D-G); however, % Oc.S/BS was significantly decreased with MIA-treatment ($p=0.03$; 95%CI [1.9 to 46.9]) (Fig.3A-B) while CTSK-positive stain remained unaltered with MIA-treatment (Fig.3D-G). Further statistical analysis revealed a significant MIA-treatment and genotype interaction effect ($p=0.0087$; 41.1% of total variation) (Fig.3C). This shows that in DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice, bone resorption is uncoupled from cartilage degeneration in TMJOA and is unaffected in the presence of MIA-treatment, consistent with observations that MIA induces TMJOA in an osteocytic MMP13-dependent manner (Fig.3-D).

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Reduced Osteocytic-MMP13 expression in DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ is maintained in MIA-induced TMJOA.

To confirm the effects of TMJOA from suppressed MMP13 in osteocytes, we conducted IF targeting MMP13, which is visually identified in red in contrast to a Rabbit IgG negative control (Fig.4A). As expected, results reveal a significant reduction in osteocytic MMP13 in saline DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ TMJs compared to saline Controls (subchondral bone: p=0.01; 95%CI 1.5 to 15.0]) (Fig.4B-E). Surprisingly, MIA-treated Controls also have significantly reduced MMP13 expression in both cartilage (p=0.05; 95%CI 0.03 to 12.1]) and subchondral bone (p=0.04; 95%CI 0.4 to 12.7]), suggesting a more complex role of MMP13 in TMJOA that is inherently different from that of the knee.

TMJ Subchondral Bone Mineralization is Unaffected by MIA Treatment

We have observed that MIA induces changes to the articular surface and LCN homeostasis in the TMJ, and that osteocytic MMP13-deficiency protects against these changes. To determine the effect of these conditions on subchondral bone quality, TMJs were evaluated using microCT (Fig.5, Supplementary Table 1). MicroCT showed no effects of MIA treatment on the TMJ, regardless of the genotype (Fig.5B-I). For all bone parameters, including bone mineral density (BMD), bone volume fraction (BV/TV), connectivity (Conn) density, trabecular number (Tb.N), tissue mineral density (TMD), trabecular thickness (Tb.Th), trabecular spacing (Tb.Sp), and structural model index (SMI) (Fig.5B-I), no discernible differences were detected across genotype or treatment.

Injury Protective Effect of Osteocytic MMP13-Deficiency on Subchondral Bone

While TMJ phenotypes of saline-injected Cre-negative control and DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice were identical, DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ knees had more severe OA at baseline than Cre-negative controls[3]. However, it is unclear whether DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ knees also differed

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from the DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ TMJ in their protection from injury-induced OA. In our prior studies, surgical injury to DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ knees exacerbated the loss of subchondral bone osteocyte canalicular length, but had no effect on knee Mankin score or subchondral trabecular bone volume[3]. Therefore, we further examined the effect of injury on DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ knees radiographically and histologically. Consistent with prior reports[4, 40], MLI of the knee significantly increases SBP thickness in the femur of Cre-negative control mice (Fig.6A), with the most profound effects on the injured medial side (anterior: p=0.01; 95%CI [-64.8 to -5.8], posterior: p<0.0001; 95%CI [-76.7 to -25.2]) (Fig.6D,E). Notably, SBP thickness was resistant to injury in DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ knees (Fig.6A-E). The lateral side of the knee showed no significant differences in any condition (Fig.6A-C). We further examined the effect of knee injury on osteocyte lacunar density and found a mild increase in injured DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ knees, but overall no significant differences based on genotype or injury status (Fig.6F-G). Therefore, despite differences in baseline DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ knee and TMJOA phenotypes, both joints show some protection from injury-induced OA in the setting of osteocytic MMP13-deficiency.

DISCUSSION

In this study, we investigated whether osteocytic MMP13 plays a causal role in the progression of joint degeneration of the TMJ, as it does in the knee[3]. We previously reported that osteocytic deficiency in MMP13, an enzyme required for PLR[12], is sufficient to cause cartilage degeneration in the knee. In contrast, we found that the TMJ of DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice did not differ from Cre-negative controls in baseline saline-injected conditions, suggesting that the homeostatic effects of osteocytic MMP13 are joint-specific. Although MIA-induced TMJ injury of Cre-negative control mice caused degeneration of TMJ cartilage and some changes in subchondral bone (e.g. increased lacunae density of the LCN), MIA-injected DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice show no significant changes in cartilage, LCN, collagen organization, TRAP

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or CTSK expression when compared with their Cre-negative saline or MIA-challenged littermates. Comparing these findings with those of injured knees shows that DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ SBP is also resistant to injury-induced changes. Although ablation of osteocytic MMP13 prevented the MIA-dependent increase in TMJ lacunar density, no significant differences were observed in knee joint lacunar density. This was unexpected since we previously reported shortened canalicular length and altered collagen organization in DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ knees[3], suggesting that lacunar density and canalicular length are regulated independently in a joint-specific and/or injury-specific manner. Therefore, this study concludes that for both the TMJ and the knee, osteocytic MMP13-deficiency protects against thickening of subchondral bone in response to joint injury.

The loss of cellularity in the MIA-treated TMJ articular cartilage prompted unbiased MALDI-MSI analysis of the TMJ to track changes in chondrocyte differentiation and ECM remodeling (Supplemental Fig.2). Application of MALDI-MSI to bone and other mineralized tissues is technically challenging. Indeed, poor adhesion of demineralized TMJ tissue to the slides limits our ability to make quantitative conclusions. Nonetheless, analysis of 1 TMJ for each of the 4 conditions in this study identifies the 6 most abundant proteins detected by MALDI-MSI, including collagen alpha-1(I) (COL1A1, 65%) and collagen alpha-1(II) (COL1A2, 24.7%), with lesser amounts of collagen alpha-1(III) (COL3A1), aggrecan (ACAN), elastin (ELN), elastin microfibril interfacer 1 (EMILIN-1) (Supplemental Fig.2A). Structural features in the photomicrograph (Supplemental Fig.2B) aid in visualizing the heuristic proteomic segmentation of the top 6 peptides, revealing qualitative differences between the vehicle and MIA-treated groups (Supplemental Fig.2C). Segmentation of 2 peptides, COL1A1 domain [572-583] (Supplemental Fig.2D) and COL1A2 domain [565-582] (Supplemental Fig.2E), distinguish the TMJ condyle in MIA-treated from vehicle-treated groups and may also correspond to genotype-dependent differences in TMJOA. The ECM proteins that appear to be

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enriched in MIA-induced TMJOA have been implicated in other proteomic studies of OA, with increased COL1A1, COL3A1, and ACAN during OA progression in humans[41, 42] and horse[43]. Although conclusions are limited by low sample size, these encouraging results motivate the development of custom TMJ peptide libraries to identify and track protein localization throughout the progression of TMJOA.

Our results suggest that the role of osteocytic MMP13 is region-dependent. The TMJ and knee joints differ molecularly, structurally, and functionally. These differences occur during development, as cells derived from cranial neural crest form the TMJ via intramembranous formation, whereas mandibular condyle forms through endochondral ossification[44-46]. The anatomy of fully developed joints supports unique functional and mechanical demands, such that the meniscus in the knee joint manages compressive loading in locomotion[47, 48], and the TMJ articular disc manages shear forces during for eating and speech[44, 49, 50]. The articular cartilage of the knee joint is composed of hyaline cartilage consisting of collagen type II, in contrast to fibrocartilage consisting of collagen type I covering the TMJ[44]. Finally, as opposed to the primary function as articular cartilage in the knee joint distinct from the epiphyseal growth plate, the TMJ condylar cartilage served both as a hybrid articular and growth site. Here, we observe no effect of osteocytic MMP13 on collagen organization in TMJ subchondral bone, unlike the knees. At baseline, knees of DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice show more joint degeneration relative to their Cre-negative littermates, but DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ TMJ tissues are intact. These differences in baseline TMJ and knee joint mechanics and cartilage composition may propagate to differences in OA progression at each site. Despite the apparent reduction in MMP13 in MIA-treated control TMJs, our data suggest that osteocytic MMP13 is not necessary for TMJ homeostasis but is necessary to drive TMJOA. Indeed, we observe that osteocytic MMP13-deficiency protects from MIA-induced OA in the TMJ, but exacerbates knee OA[3]. The current studies on the role of osteocytic MMP13 in joint homeostasis adds

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cellular and molecular detail to our understanding of mechanisms responsible for differences between the TMJ and the knee. Nonetheless, additional research will be needed to understand the joint-specific effects of osteocytic MMP13.

With respect to bone/cartilage crosstalk, alterations in the subchondral bone can contribute to secondary changes in the cartilage and perpetuation of OA[52-54, 55]. Although ablation of MMP13 in osteocytes was confirmed by IF in our DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ model (Fig.4), the significant downregulation of MMP13 expression in the condylar cartilage of DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ was unexpected, as we previously reported unaltered MMP13 expression in knee articular cartilage of these mice[3]. Our data indicate no evidence that chondrocyte differentiation is the underlying driver of TMJOA differences observed; however, MIA does induce chondrocyte apoptosis[51]. The low levels of MMP13 in the condyle cartilage may be a secondary effect of osteocyte-specific ablation of MMP13, or off-target effects of DMP1-cre in the mandibular condyle chondrocytes. Among possible mechanisms, avascular knee cartilage may be more dependent on an intact LCN to provide indirect vascular support across the robust bone/cartilage interface in the knee. If so, LCN degeneration due to PLR suppression in DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ subchondral bone may contribute to cartilage degeneration in the knee[3]. On the other hand, the more integrated bone/cartilage interface of the TMJ may allow subchondral vasculature to support the avascular TMJ cartilage, independently of osteocytic PLR suppression and LCN integrity. The unique bone/cartilage interfaces of the TMJ and knee may also contribute to joint-specific differences in inflammatory biomarkers, such as cartilage oligomeric matrix protein (COMP) and prostaglandin E_2 (PGE_2), in human synovial fluid from TMJ or knee OA patients[56,57]. Permeability studies are needed to test this hypothesis, using animal models that replicate healthy and diseased joint crosstalk to evaluate transport across the bone/cartilage interface in TMJ and knee joints[58].

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To induce OA in the TMJ, in this study we injected monosodium iodoacetate (MIA) into the mouse TMJ. MIA causes OA by inducing chondrocyte death[51], in contrast to the surgery-induced MLI-injury model of OA in the knee. Despite these differences, DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mouse knee and TMJs resisted the injury-induced cartilage degeneration and other changes observed in their Cre-negative littermate controls. The increased osteocyte number, elevated TRAP, and modest CTSK expression in MIA-treated control TMJs may contribute to cartilage degeneration, whereas the lack of change in DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ TMJs may preserve joint homeostasis even following MIA. Although MIA-treatment of rats and mice has been shown to induce lesions in TMJ cartilage and bone[23, 36], microCT assessment revealed no TMJ subchondral bone changes across any group in this study. Differences in the TMJ bone response to MIA-treatment may be due to species or background strain-specific responses, MIA dosage, or differences in the microCT region of interest, since subchondral bone was evaluated in this study, rather than the entire TMJ condyle.

It is well known that mesenchymal progenitor cell-derived MMP13 negatively regulates bone mass, with the absence of MMP13 in limb mesenchymal cells resulting in increased bone mass[59] while, chondrocytic MMP13 contributes to cartilage degradation and inhibiting MMP13 reduces osteoarthritis progression[60, 61], however, the role of osteocytic MMP13 in TMJOA remains unknown, which the current study aims to address. Although the definitive mechanism of how osteocytic MMP13 exacerbates TMJOA remains unclear, possible mechanisms may involve spatial regulation of MMP13 expression and activity, integrity or porosity of the cartilage/subchondral bone interface, or differences in shear or compression loading between the TMJ and the knee, since MMP13 expression is mechanically responsive[47, 48]. This novel study reveals osteocytic-MMP13 as a potential therapeutic target to mitigate TMJOA; however, further studies are required to uncover the responsible cellular mechanism.

By investigating osteocytic MMP13 in the TMJ and in the knee, this study further supports a role for subchondral bone osteocytes in joint homeostasis and in the progression of OA in both the knee and TMJ. The role of osteocytic MMP13 in regulating joint homeostasis is distinct in the knee and TMJ in baseline conditions, but in both cases, the loss of osteocytic MMP13 mitigates the severity of injury-induced OA in these joints. Specifically in the TMJ, osteocytic MMP13-deficiency may protect against MIA-induced cartilage degeneration, LCN, and elevated osteocytic CTSK and TRAP expression.

We recognize some limitations in this study. Although microCT settings were optimized for each joint, the TMJ may require a higher resolution scan, such as synchrotron-generated CT, to increase sensitivity. Though power analysis was based on prior studies[3, 4, 12, 18, 23, 26, 33]; a larger sample size will further improve the ability to discriminate the effect of each variable on the subchondral bone phenotype and understand its relationship to the LCN in the TMJ and the knee. This study was powered to detect significant osteocyte MMP-13-dependent differences in the effects of MIA-induced injury on TMJ degeneration (Fig.1B) and osteocyte lacunar number (Fig.2B), but additional statistical power is needed to identify peptides with MMP-13 and injury-dependent differences in expression levels or localization (Supplemental Fig.2B). While further examination is needed, this study advances a better understanding of TMJ cartilage and subchondral bone interactions in the progression of TMJOA.

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AUTHOR CONTRIBUTIONS

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Cristal S. Yee: analysis and interpretation of the data, drafting of the article, technical support, collection and assembly of data

Marianne Demirdji: analysis and interpretation of the data, collection and assembly of data, TMJ MIA model development and characterization

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Alena Larios: analysis and interpretation of the data, collection and assembly of data

Christoforos Meliadis: analysis and interpretation of the data, collection and assembly of data

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Sunil Kapila: analysis and interpretation of the data, conception and design, critical revision of the article for important intellectual content, final approval of the article, obtaining of funding

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ROLE OF THE FUNDING SOURCE

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CONFLICT OF INTEREST

The authors do not have any conflict of interest to disclose.

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Journal Pre-proof

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DATA AVAILABILITY STATEMENT

The datasets generated for this study are included in the article/Supplementary Materials. Further inquiries can be directed to the corresponding authors.

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DISCLAIMER

This research was supported in part by the Intramural Research Program of the National Institutes of Health (NIH). The contributions of the NIH author were made as part of their official duties as NIH federal employees, are in compliance with agency policy requirements, and are considered Works of the United States Government. However, the findings and conclusions presented in this paper are those of the authors and do not necessarily reflect the views of the NIH or the U.S. Department of Health and Human Services.

FIGURE LEGENDS

Figure 1. Osteocyte deficiency of MMP13 protects the TMJ from MIA-induced joint degeneration. Safranin O / Fast Green stain on TMJs (n=5-6) shows loss of proteoglycan staining (red) in the cartilage, a hallmark of cartilage degeneration, in MIA-treated control, but not in DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice (**A**, 40X, scale bar = 130µm). Baseline TMJ phenotypes between the two saline-treated genotypes shows no differences in Modified TMJ Mankin Scores **(B-F)**. Although MIA significantly increases all variables comprising the Modified TMJ Mankin Score **(C-F)** and the total score in Cre-negative control cohorts **(B)**, DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice do not show any MIA-induced changes in Modified TMJ Mankin Scores **(B-F)**, revealing protection of cartilage in the DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ mice. Data are presented as mean + SD. Statistically significant differences ($p<0.05$) were determined by two-way ANOVA with post-hoc Tukey.

Figure 2. Osteocytic MMP13-Deficiency Prevents Injury-induced Changes to TMJ Subchondral Bone LCN without Altering Collagen Organization. LCN are visualized by Ploton Silver Stain of the TMJ (n=5-6) **(A**, 100X, scale bar = 30µm). Lacunar density in the TMJ condyle is significantly higher in MIA-treated Cre-negative mice compared to saline

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controls, but is unaffected by MIA in the DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ TMJ **(B)**. Collagen orientation in the subchondral TMJ was visualized from Picrosirius Red Stain **(C)** and analysis of collagen organization **(D)**, maximum distribution of orientation **(E)**, and full width half maximum (FWHM) **(D)** were unaltered in all groups (n=5) (40X, scale bar = 80μm). Data are presented as mean + SD. Statistically significant differences ($p<0.05$) were determined by two-way ANOVA with post-hoc Tukey.

Figure 3. Maintenance of Bone Resorption and Cathepsin K expression following injury in the DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ TMJ. TRAP staining shows resorption activity (**A**, N=4, 20X, scale bar = 200μm) primarily in the subchondral bone regions. The effect of MIA on TRAP activity, evaluated as the percentage of osteoclast surface over bone surface (Oc.S/BS), in Cre-negative control mice is opposite that in DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ TMJs (**B**) with ANOVA results indicating a significant genotype and treatment interaction effect (**C**). IHC using anti-cathepsin K (CTSK) shows increased CTSK intensity (**D**, N=5, 40X, scale bar = 100μm) and trends of increased CTSK-positive staining overall, and in articular cartilage and in subchondral bone regions of MIA-treated Cre-negative control mice **(E-G)**. Basal levels of CTSK expression are maintained in DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ TMJs treated with MIA **(E-G)**. Data are presented as mean + SD. Statistically significant differences ($p<0.05$) were determined by two-way ANOVA with post-hoc Tukey.

Figure 4. Osteocytic MMP13 expression is reduced in MIA-induced TMJOA. Immunofluorescence on TMJ sections using negative control (Rabbit IgG) (**A**, 20X, scale bar = 100μm; **a1**, 40X, scale bar = 100μm) or anti-MMP13 (**B**, 40X, scale bar = 100μm). The dotted white line indicates the borderline between the cartilage and subchondral bone. Results show total MMP13 expression (**C**), which includes cartilage (**D**) and subchondral bone (**E**) regions of the TMJ. While MIA-treated Control mice had decreased MMP13 expression (**C**), MMP13 levels in MIA-treated DMP1-cre$^{+/-}$;MMP13$^{fl/fl}$ did not change from DMP1-cre$^{+/-}$

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;MMP13$^{fl/fl}$ basal levels (**C-E**). Data are presented as mean + SD. Statistically significant differences ($p<0.05$) were determined by two-way ANOVA with post-hoc Tukey.

Figure 5. TMJ Subchondral Bone is Unaffected by MIA Treatment. µCT was used to generate and quantify 3D reconstructed images of the TMJ condyle surface (**A,** scale bar = 200µm), revealing that MIA-treatment does not affect the subchondral bone of the TMJ in the presence or absence of osteocyte-specific MMP13 **(B-I)**. No changes across genotype and treatment groups are observed in any bone parameters (n=4-5/group): bone mineral density (BMD) **(B)**, bone volume fraction (BV/TV) **(C)**, connectivity density (Conn Density) **(D)**, trabecular number (Tb. N) **(E)**, tissue mineral density (TMD) **(F)**, trabecular thickness (Tb.Th) **(G)**, trabecular spacing (Tb.Sp) **(H)**, and structural model index (SMI) **(I)**. Data are presented as mean + SD. Statistically significant differences ($p<0.05$) were determined by two-way ANOVA with post-hoc Tukey.

Figure 6. Osteocytic MMP13-deficiency Prevents Injury-Induced Knee SBP Thickening. Cre-negative control mice with MLI-injury show thicker SBP in the injured medial region as visualized by 3D reconstructed images of the femoral knee (**A**, scale bar = 200µm) and verified quantitatively (n=10-11) **(D,E)**. However, osteocytic MMP13-deficiency mitigates injury-induced SBP thickening **(D,E)**. No difference in SBP thickness in the lateral side is observed **(B,C)**. LCN visualized by Ploton Silver Stain of knee subchondral bone (n=5) **(F)** (100X, scale bar = 30µm) shows no significant effects on lacunar density due to genotype or injury (n=6) **(G)**. Data are presented as mean + SD. Statistically significant differences ($p<0.05$) were determined by two-way ANOVA with post-hoc Tukey.

Table 1: Modified Mankin Score to quantify TMJOA.

1. Cartilage Erosion	Presence of an intact, eroded, and/or fibrillation on the surface of the condylar cartilage.

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	a. Smooth non-eroded cartilage	0
	b. Rough non-eroded cartilage	1
	c. Superficial fibrillation	2
	d. Separation of uncalcified from calcified cartilage	3
	e. Erosion of uncalcified cartilage only	4
	f. Erosion extending into calcified cartilage	5
	g. Erosion down to subchondral bone	6
2. Chondrocyte periphery staining	Intensity in Safranin O stain in the surrounding region of the chondrocytes.	
	a. Normal	0
	b. Slightly enhanced	1
	c. Intensely enhanced	2
3. Spatial arrangement of chondrocytes	Diffuse hypercellularity was noted as a clear increase in the general number of chondrocytes, while clustering was agreed upon as three or more cells congregated together. Hypocellularity was chosen when the cartilage had areas lacking any chondrocytes.	
	a. Normal	0
	b. Diffuse hypercellularity	1
	c. Clustering	2
	d. Hypocellularity	3
4. Background Staining Intensity	Intensity of the counterstain, Fast Green.	
	a. Normal	0
	b. Slight reduction	1
	c. Moderate reduction	2
	d. Severe reduction	3
	e. No dye noted	4
5. Subchondral bone deterioration	Anatomy of the subchondral bone as identified with Fast Green staining adjacent to the cartilage (Safranin O stain) to observe erosion and sclerotic changes.	
	a. Normal	0
	b. Local bone defect: single/superficial	1
	c. Local bone defect: multiple/single deep	2
	d. Bone sclerosis	3

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6. Demarcation	The boundary between the cartilage and the subchondral bone. Examples of a well-demarcated versus a vaguely demarcated one are shown below. The sample that is not demarcated has finger-like projections from the cartilage infiltrating into the bone.	
	a. No	0
	b. Yes	1

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Figure 1

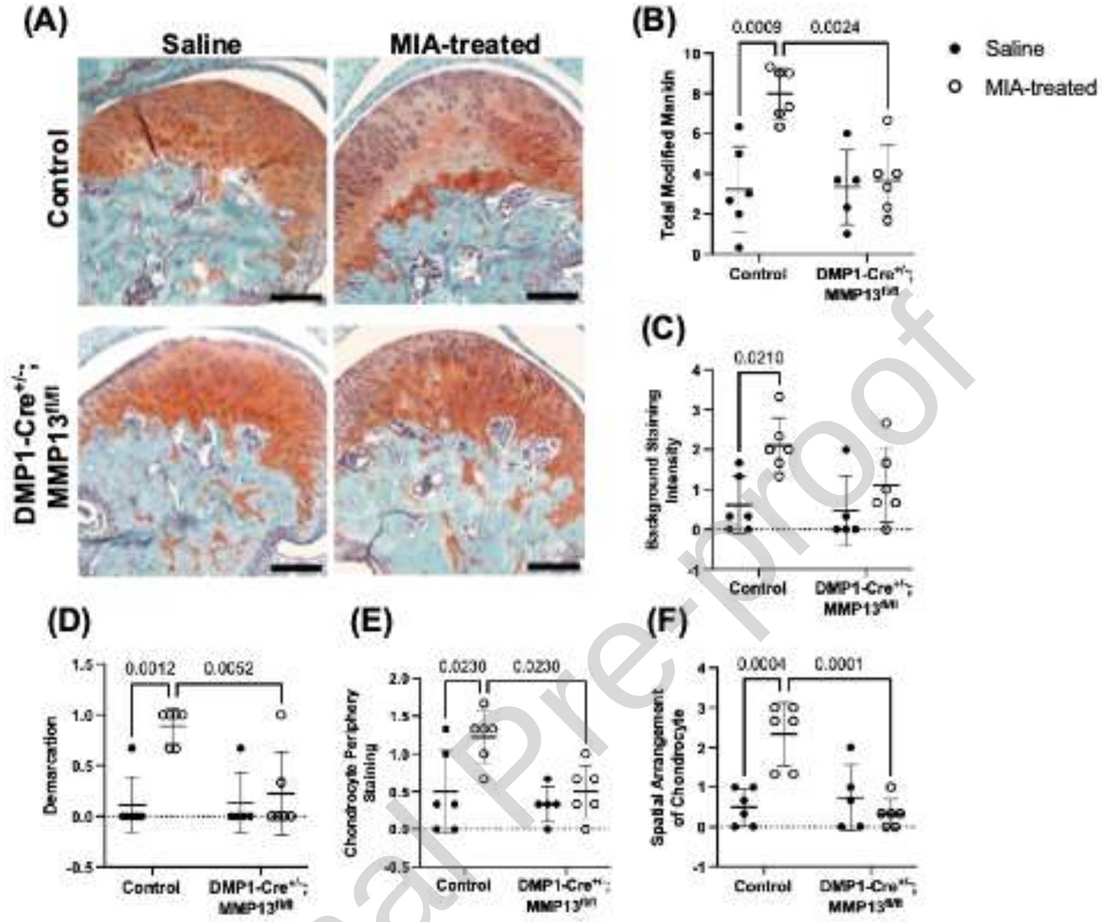

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Figure 2

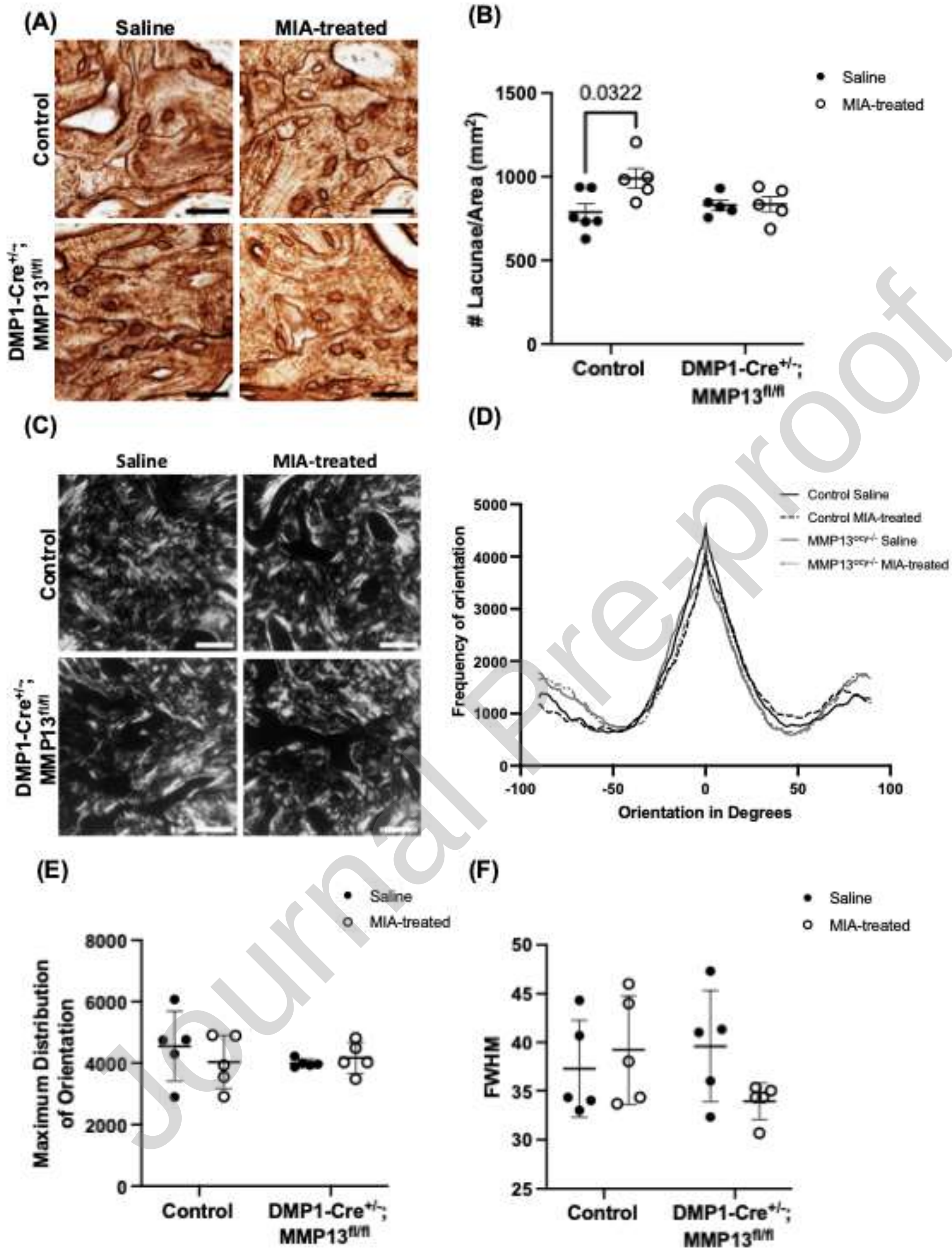

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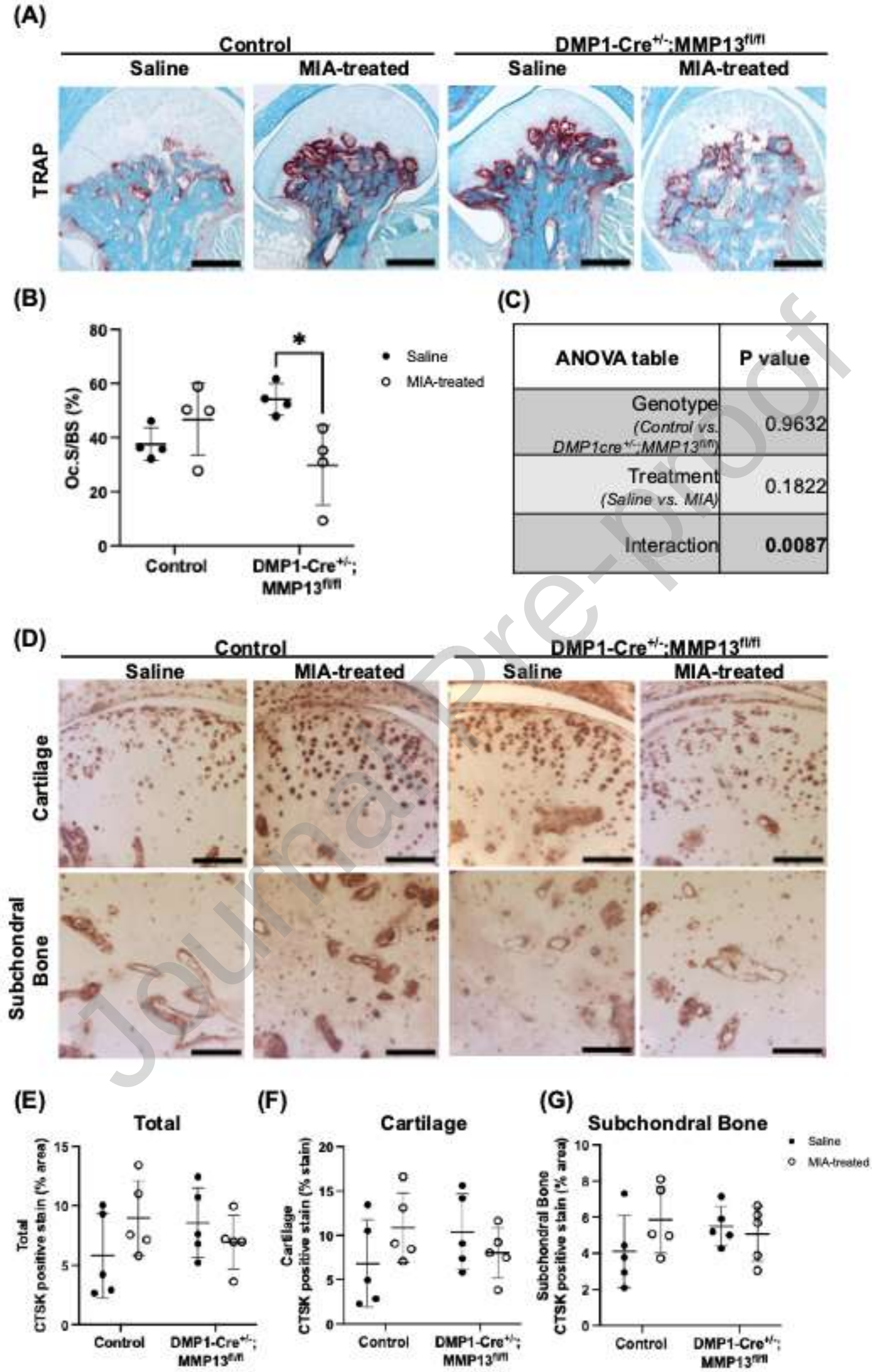
Figure 3



ANOVA table	P value
Genotype (Control vs. DMP1cre+/-;MMP13fl/fl)	0.9632
Treatment (Saline vs. MIA)	0.1822
Interaction	**0.0087**

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Figure 4

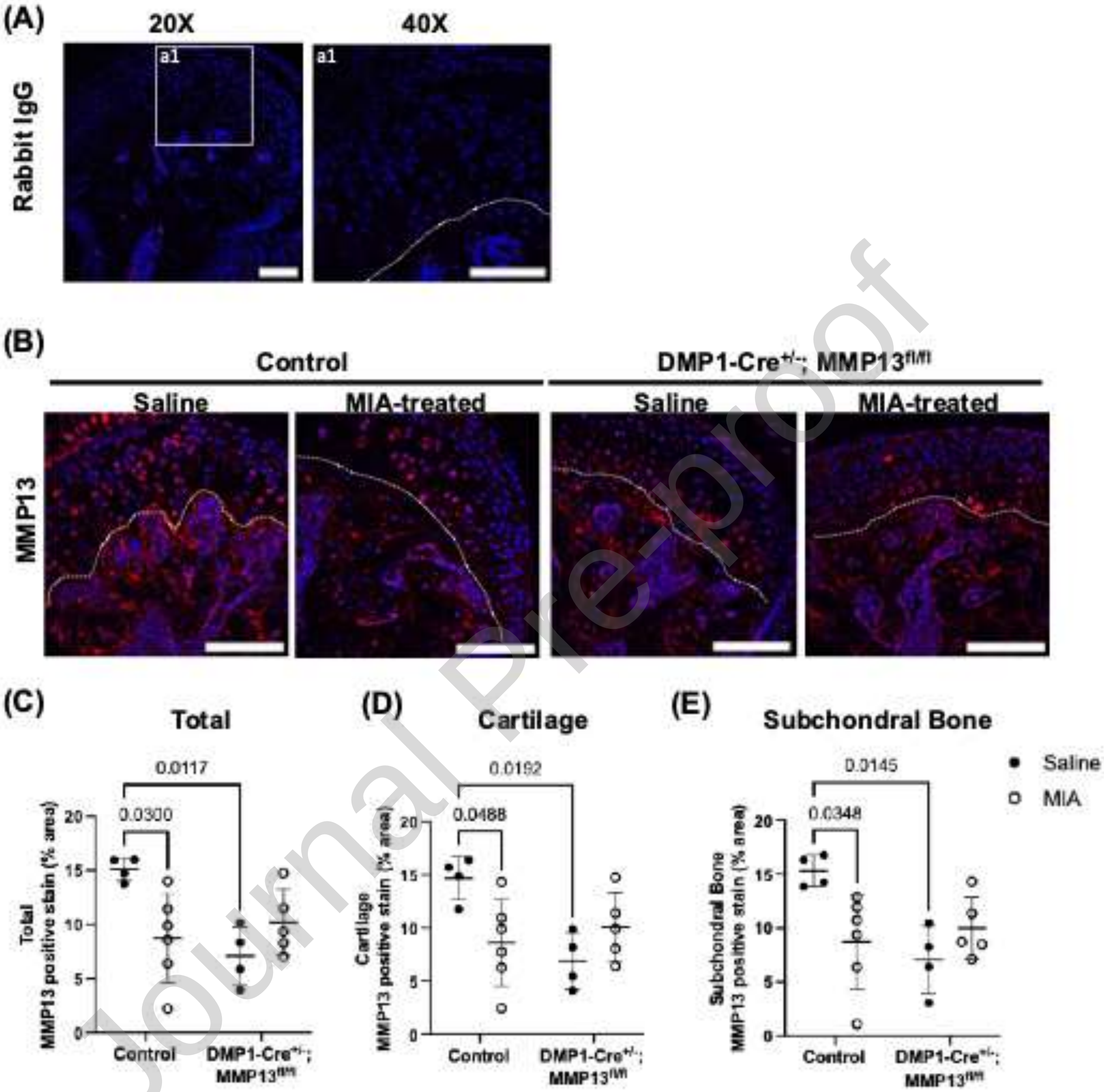

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Figure 5

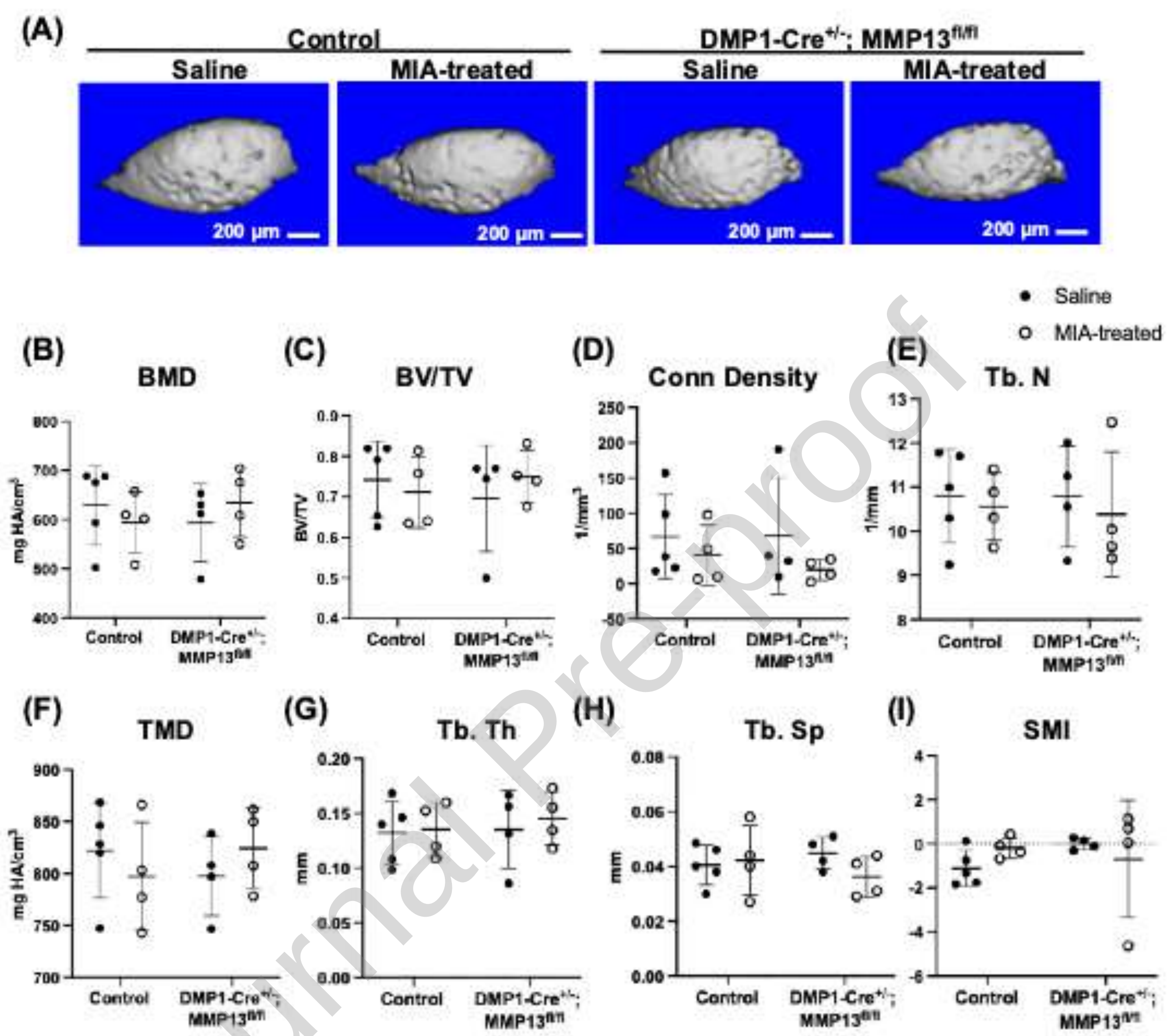

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Figure 6